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First palaeomagnetic results from the Cycladic Massif, Greece, and their implications for Miocene extension directions and tectonic models in the Aegean

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Abstract

The Aegean region is characterised by major fault-bound blocks with markedly contrasting structural trends. Recent models account for these variations either through differential block rotation or through a process of radial extension. The Cycladic Massif in the back-arc region to the Hellenic trench represents a critical area where apparently divergent extension directions are observed. This contribution provides the first palaeomagnetic results from this area. Magnetic remanence and AMS data are presented for Middle Miocene granodiorites on the islands of Mykonos and Naxos. These rocks are exposed in the footwalls of major extensional detachments within the metamorphic core complexes that characterise the Cycladic Massif. A clear correlation between the orientation of an L–S fabric developed within the intrusions and the magnetic fabric data is observed, demonstrating the presence of a 'normal' magnetic fabric of extensional origin. The mean extension directions inferred from the orientations of the maximum axes of the magnetic susceptibility ellipsoids are markedly divergent at 064° (Mykonos) and 351° (Naxos).

Northeasterly in situ magnetic remanences are observed on Mykonos, whereas those on Naxos are directed to the NNW. Neither direction corresponds to the appropriate palaeomagnetic reference direction for the Miocene. Standard palaeomagnetic tilt corrections cannot be applied at these sites because of the lack of palaeohorizontal markers. Deviations of in situ remanences away from the reference direction are potentially due to rotation around any one of an infinite number of net tectonic rotation axes. Each of these may be arbitrarily decomposed into components of tilting and vertical axis rotation. Geologically realistic solutions which restore the in situ magnetisations back to the reference direction may be found by consideration of regional structural trends, assuming that tilting is controlled by the major block-bounding faults. These solutions indicate significant components of differential vertical axis rotations across the Cyclades. Mykonos is bound by NW–SE trending regional faults and is shown to have rotated clockwise by ca. 22°. In contrast, Naxos is bound by ENE–WSW trending structures and has rotated anticlockwise by ca. 33°. The apparent divergence of extension directions inferred from both the magnetic anisotropy and structural data is resolved by correcting the mean AMS data for subsequent rotation using the rotation solutions derived from the magnetic remanence data. The data imply that the primary Middle Miocene extension direction in the Cycladic Massif was nearly uniform along a NNE–SSW azimuth.

We consider that the uniform extension direction and opposing senses of block rotation determined for this portion of the

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Cycladic massif are most consistent with a recent model describing deformation within the Aegean in terms of a system of broken, fault-bounded slats. Within this scenario, Mykonos occupies a position on a clockwise rotating slat which continues northwest, at least as far as Evvia. A major ENE–WSW trending fault between the islands, previously identified on shallow seismic sections, separates this domain from an anticlockwise rotating slat which includes Naxos.

Keywords: Miocene; paleomagnetism; rotation; tectonics; Cyclades; Aegean Sea

1. Introduction

The Aegean region is one of the most rapidly deforming parts of the Tethyan (Alpine–Himalayan) collisional system and, as such, is the focus of continued interest. It is an ideal natural laboratory for modelling present-day, active collisional tectonics and back-arc processes [1]. Tectonic models for earlier periods derived from field-based measurements may be compromised, however, by undetected reorientations of observed structures (e.g. by vertical axis rotations). In such a dynamic region this problem may even apply to models for the neotectonic period.

The Aegean is characterised by widespread extension associated with subduction along the Hellenic trench (Fig. 1, insert). The subduction zone is believed to have shown a progressive increase in curvature since initiation in the Miocene [2], and many attempts have been made to model the pattern of

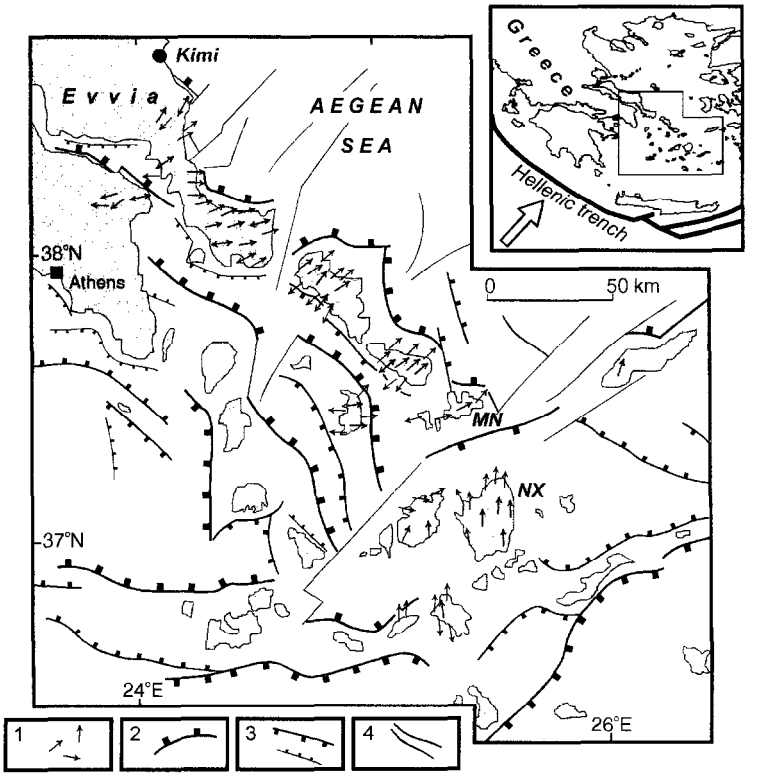


Fig. 1. Regional structural compilation of Gautier and Brun [4], showing extension directions and main faults across the Aegean archipelago. Note major deviation in extension directions within the Cycladic massif between Mykonos and Naxos Islands. *MN* = Mykonos Island; *NX* = Naxos Island; *1* = ductile shear sense indicators within metamorphic core complexes and Miocene intrusions; *2, 3, 4* = neotectonic faults; *2* = major normal faults; *3* = second-order normal faults; *4* = strike-slip and/or transfer faults. Modified from Gautier and Brun [4], fig. 4.

extension in the back-arc region within this framework (e.g. [3,4]). In each case, patterns of regional extension have been reconstructed through analysis of fault plane and shear zone fabric orientation data without correction for the effects of subsequent tectonic rotations. Existing palaeomagnetic data from around the Hellenic arc [5,6] and within the internal Aegean zones [7–10], together with models based on earthquake focal mechanisms [3], suggest such rotations have played an important role in the neotectonic development of the Aegean. In particular, the Cycladic Massif of the southern Aegean is characterised by major fault-bounded blocks with markedly contrasting structural trends. Extension directions derived from field measurements in this critical area are typically consistent within individual blocks, but show differences of up to 70° between adjacent blocks [4]. Several studies account for these variations in terms of systematic regional changes in the primary extension direction [11–13]. Furthermore, such patterns are considered to be representative of extension directions in the back-arc region throughout the Lower Miocene to early Quaternary period [4]. Although these models incorporate the suggestion that tectonic rotations may have occurred, the absence of palaeomagnetic data has so far prevented evaluation of the effect of rotation on measured extension directions.

This contribution provides the first palaeomagnetic data from the Middle–Late Miocene intrusive rocks of the Cycladic Massif. The lack of previous studies is in part due to the absence of palaeohorizontal control on such plutonic rocks, which contain no palaeohorizontal indicators and which are overlain tectonically by Tertiary sediments. We demonstrate that tectonically useful palaeomagnetic data can be obtained in such settings, and use these data to restore structurally derived Neogene extension directions [4,14,15] to a pre-rotation framework. The implications of this reconstruction for existing tectonic models of the Aegean back-arc region are discussed.

2. Geological setting

The Aegean domain has been dominated by widespread extension since at least the Middle Miocene [16,17]. Extensional tectonics are superimposed on crustal material previously thickened during the Mesozoic–Cenozoic Hellenic orogeny. Although occupying a back-arc position above the northerly dipping Hellenic subduction zone, extensional tectonics are thought to reflect a combination of collapse of the over-thickened crust and roll-back of the subducting slab [13]. HP/LT metamorphic rocks exposed throughout the Cycladic Massif are thought to have been exhumed during this extension, with structural and metamorphic characteristics typical of Cordilleran-type metamorphic core complexes [18]. Middle–Late Miocene granitoids were intruded during the later stages of extensional exhumation behind the Hellenic trench [15]. This study focuses on two such granodioritic intrusions exposed on Mykonos and Naxos islands.

The ca. 11 Ma granodiorite on Mykonos [19] occupies a position in the footwall of a major low-angle extensional detachment [15] and displays a pronounced L–S tectonite fabric. A well developed NE dipping mylonitic foliation and an ENE trending mineral elongation lineation have previously been documented [15]. Kinematic indicators are consistent with a top-to-the-NE sense of shear along the major detachment fault [15].

Structural relationships on Naxos [14] are similar to those observed on Mykonos [15]. Intrusion of a 13–11 Ma granodiorite [20,21] post-dates peak metamorphism in the core complex [14], and is again considered to be coeval with extensional exhumation. Early deformation of the granodiorite produced a mylonitic foliation dipping shallowly to the NW with a N-NNW trending mineral elongation lineation [14,22]. These are overprinted by a subparallel cataclastic foliation and associated high-angle normal faults [22]

3. Sampling and methodology

Samples were drilled in situ at 12 sites using a standard rock drill and were oriented with both magnetic and sun compasses. Two sites in the Naxos granodiorite that were extensively chloritized, were found to have immeasurable magnetizations and are not discussed further. The orientation of elongate

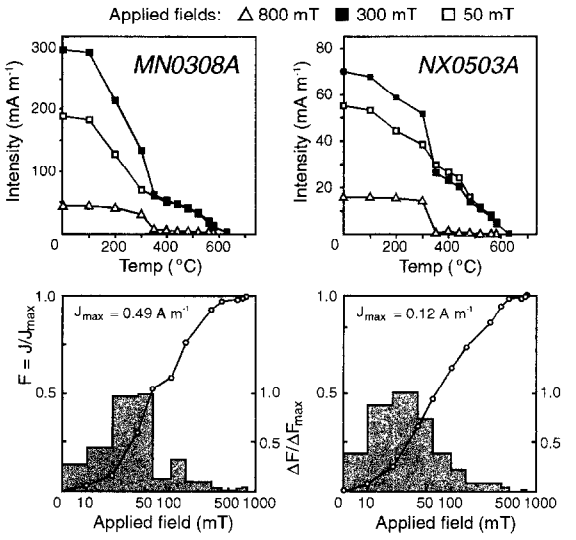


Fig. 2. IRM acquisition curves showing the presence of low to medium coercivity magnetic grains.

enclaves, mineral elongation lineations and foliation planes were recorded where visible.

Although the intrusions are overlain by Tertiary sediments, these were emplaced tectonically and could not be used to provide palaeohorizontal control. Results are therefore presented in in situ (geographic) coordinates. The lack of palaeohorizontal control has been overcome by modelling the effects of varying amounts of structural tilt on the in situ palaeomagnetic results.

Magnetic remanence measurements were made using a CCL superconducting magnetometer, and their quality assessed statistically [23]. Both thermal and alternating field (AF) demagnetization techniques were employed. Characteristic components of magnetization were found using orthogonal vector plots and principal component analysis [24] and site mean remanence directions computed using Fisherian statistics. The bulk susceptibility and anisotropy

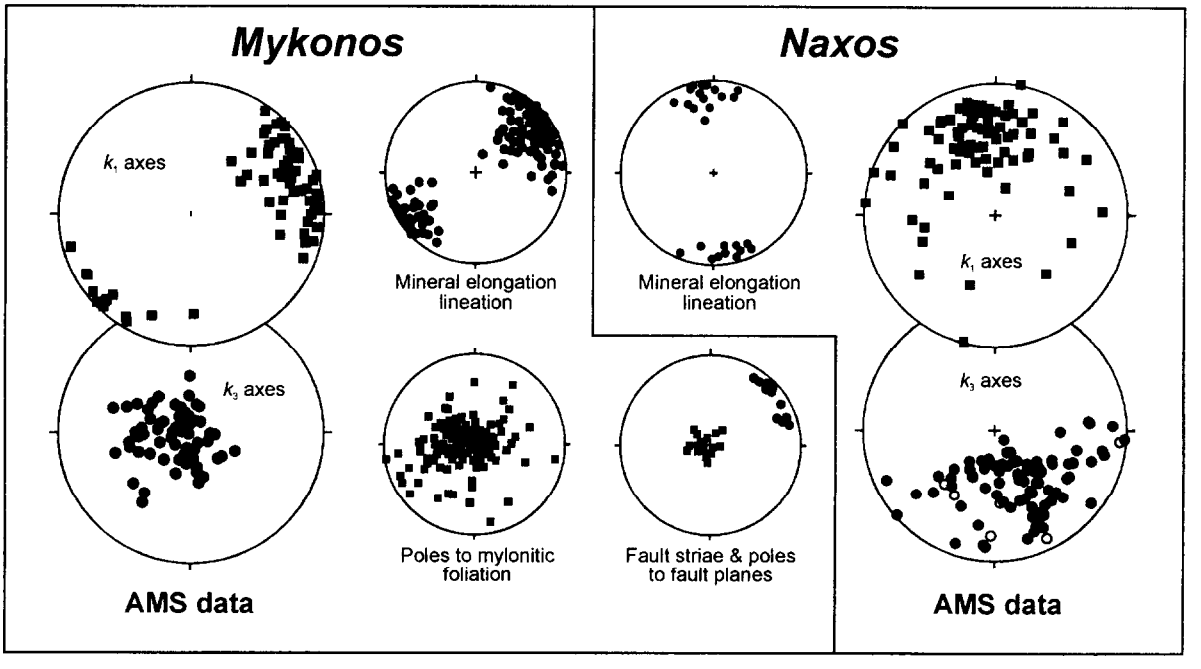


Fig. 3. Comparative stereoplots of structural field data [14,15] and anisotropy of magnetic susceptibility (AMS) data [this study]. The close agreement between petrofabric and AMS data suggests the presence of a normal magnetic fabric of extensional origin. Maximum (k_1) axes of magnetic susceptibility can be used with confidence, therefore, as a proxy for structural lineation data.

of magnetic susceptibility (AMS) of all samples were measured using a Minisep anisotropy meter. Anisotropy of isothermal remanence was also measured for selected samples [25].

4. Magnetic properties

Isothermal remanent magnetization (IRM) acquisition curves rise steeply and appear to reach saturation by fields of 0.3–0.8 T (Fig. 2). Histograms of the rate of change of IRM acquisition suggest that low to medium coercivity minerals are dominant. Unblocking temperatures associated with the hard, medium and soft coercivity fractions were identified by thermal demagnetization of orthogonal IRMs acquired in fields of 0.8, 0.3 and 0.05 T [26]. All three coercivity fractions show dramatic reductions in IRM intensity at 320°C (Fig. 2). This demonstrates the presence of pyrrhotite in a wide range of coercivities, extending from 0.05 T up to 0.8 T. The low and medium coercivity fractions are completely demagnetized at 580°C, indicating the presence of magnetite with a range of coercivities extending from 0.05 T up to 0.3 T. There is no indication of haematite or goethite. The magnetic remanence properties of these rocks are therefore due to pyrrhotite and magnetite.

5. Comparison of magnetic and petrofabrics

The magnetic fabric of the intrusions sampled was described by the orientation of AMS ellipsoids, which reflect the preferred orientation of anisotropic magnetic minerals. Corrected anisotropy degrees of individual specimens range from 1.02 to 1.17. The anisotropy of isothermal remanence ellipsoids [25] was found to be coaxial with AMS ellipsoids, but the degree of anisotropy was higher.

It is generally assumed that the AMS ellipsoid is coaxial to the petrofabric [27], with the minimum (k_3) axis perpendicular to the foliation and the maximum (k_1) axis parallel to the lineation. This assumption is only justified if supporting structural data are available, since inverse magnetic fabrics (where k_1 and k_3 axes are interchanged) are increasingly being recognised (e.g. in Tertiary leucogranites; [28]).

Table 1
Magnetic fabric and palaeomagnetic results

Site	Anisotropy of magnetic susceptibility								In situ magnetic remanence				
	n	k_{geom}	Mean orientation of k_1 axes				P_J	T	n	Dec	Inc	α_{95}	K
			Dec	Inc	α_{95}	K							
Mykonos													
MN01	12	104.6	061.9	2.1	15.4	9	1.10	0.30	6	036.8	17.7	13.0	27
MN02	16	119.9	065.0	3.1	13.6	8	1.08	0.09	6	042.8	21.8	9.2	54
MN03	15	209.5	070.7	34.1	10.1	15	1.06	0.27	6	027.8	30.5	11.3	36
MN04	13	149.7	057.2	4.0	11.0	15	1.06	−0.04	4	040.8	19.1	6.0	236
MN05	8	96.5	065.1	16.9	10.5	29	1.10	0.09	6	043.2	25.2	3.6	349
Mean	$N=5$	130.6	063.7	12.1	14.0	31	1.08	0.16	$N=5$	038.4	23.0	7.3	111
Naxos													
NX01	24	175.7	002.6	30.0	20.1	3	1.08	0.19	7	325.8	47.4	8.6	50
NX02	19	143.0	345.9	23.0	8.3	18	1.09	0.05	5	338.1	39.0	9.9	61
NX03	17	179.9	011.0	38.1	18.7	5	1.06	0.17	7	338.2	29.2	12.1	26
NX04	12	177.1	339.5	44.0	16.2	8	1.07	−0.11	7	323.7	36.9	10.7	33
NX05	15	106.1	335.8	23.9	16.9	6	1.09	−0.09	5	339.7	34.3	12.1	43
Mean	$N=5$	153.4	350.7	32.5	15.0	27	1.08	0.12	$N=5$	333.4	37.6	8.6	81

n = number of samples; N = number of sites; k_{geom} = site mean susceptibility (= antilog[(Σ log k)/n], where k = sample mean susceptibility = $\sqrt[3]{[k_1 \cdot k_2 \cdot k_3]}$); P_J = corrected anisotropy degree [41]; T = shape parameter [41]; Dec = declination; Inc = inclination; α_{95} = semi-angle of 95% cone of confidence; K = Fisher precision parameter.

Well-defined magnetic lineations and foliations (defined by clustering of k_1 and k_3 axes respectively) are observed within both intrusions. Fig. 3 shows stereoplots of data from all specimens. AMS data from Mykonos show excellent agreement with published petrofabric data [15] and our limited structural data. This agreement indicates the presence of a normal magnetic fabric and strongly suggests that the magnetic fabric on Mykonos has an extensional tectonic origin.

The less pronounced petrofabric observed on Naxos has a different orientation to that seen on Mykonos [14,22], but is again broadly coaxial to the magnetic fabric (Fig. 3). The magnetic fabric from all sites is consistent, even though the tectonic fabric could not be observed at all sites. This highlights the ability of the AMS method to detect fabrics in apparently isotropic rocks.

Specimen AMS ellipsoids vary in shape from oblate to prolate, independent of the degree of anisotropy. This reflects the variable development of the L–S tectonite fabric between individual specimens. Site mean directions of *maximum* axes of magnetic susceptibility (k_1) were calculated for each site using Fisherian statistics (Table 1), together with corresponding formation mean directions for each island. These formation means are used as a proxy for the structural lineation direction in subsequent analyses.

6. Palaeomagnetic results and interpretation

A mean NRM intensity of 0.65 mA m^{-1} was found for granodiorite specimens, compared with 1.16 mA m^{-1} for mafic enclaves within the main granodiorite bodies. Both thermal and AF demagnetisation isolated single, stable components of magnetisation. Examples are shown in Fig. 4. In general, thermal treatment proved more effective in identifying stable components of magnetisation. Maximum unblocking temperatures were in the range 320–500°C. These are equivalent to those of laboratory induced remanences, and are consistent with pyrrhotite and magnetite carriers of natural remanence.

In situ site mean directions of magnetisation are listed in Table 1. Within-site dispersions were small (K > 25), and site mean directions are shown in the

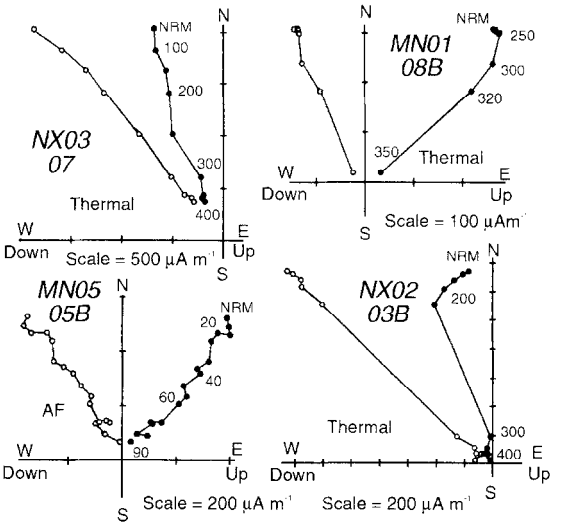


Fig. 4. Thermal demagnetisation of three-component IRMs [26], demonstrating that magnetisation is carried by pyrrhotite with an unblocking temperature of 320°C and a range of coercivities up to 800 mT, and magnetite with an unblocking temperature of 580°C and coercivities ranging from 0.05 to 0.3 T.

'clock' diagrams of Fig. 5. The characteristic magnetisation at all sites is unrelated to the present field direction in the Aegean. Remanences on Naxos are consistently directed towards the NNW, whereas those on Mykonos fall in the NE quadrant. Formation mean directions of magnetisation for each intrusion are listed in Table 1.

Kissel and Laj [6] have used palaeomagnetic data to demonstrate that the Aegean domain as a whole was linked to the African plate during most of the Tertiary. The palaeomagnetic directions reported here may, therefore, be compared to those expected from the African apparent polar wander path. The syntheses of Westphal et al. [29] and Besse and Courtillot [30] list similar African paths, which give an expected Miocene reference direction for the central Cyclades of $D = 5°$, $I = 53°$. The corresponding reference direction assuming that the Aegean was linked to Eurasia in the Middle Miocene is almost identical at $D = 7°$, $I = 53°$. We adopt the African reference direction, following the suggestion of Kissel and Laj [6], noting that the choice of reference direction makes little difference to the analyses below. The formation mean remanence directions for both the Mykonos and Naxos intrusions are significantly dif-

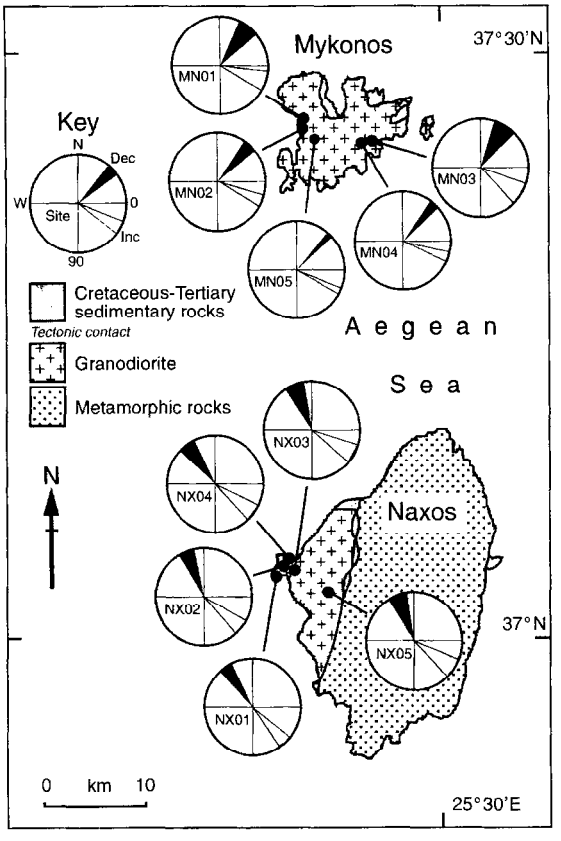


Fig. 5. 'Clock' diagrams showing the 95% confidence limits [42] associated with the mean declination and inclination at each site. All data shown in in situ (geographic) coordinates. Note the consistency of site mean directions within each sampled intrusion, and the divergence of data between islands.

ferent from this direction. In particular, observed in situ inclinations (Table 1) are substantially lower than predicted. Since no tilt corrections have been applied to these data, inclination flattenings may potentially arise from components of structural tilting. Low Tertiary inclinations have, however, been reported throughout the Mediterranean region from Iberia to Iran [29], with no generally accepted explanation [31,32]. It is more appropriate, therefore, to compare inclinations with well-constrained palaeomagnetic data from coeval formations elsewhere in the Aegean. Such data have been reported from the Kimi area of Evvia [7,9], along-strike from Mykonos, where tilt corrected inclinations of 42° are observed. Importantly, these data are derived from both intrusive and sedimentary rocks, and are supported by a positive fold test [9]. Comparing our data with a revised reference direction of $D = 005°$, $I = 42°$ gives declination rotations R and inclination flattenings F [33] for each sampled intrusion as follows:

Mykonos: $R = 33.4° \pm 6.6°$ (clockwise) $F = 19.0 \pm 6.3°$

Naxos: $R = -31.6° \pm 8.9°$ (anticlockwise) $F = 4.4° \pm 7.3°$

This reduces but does not remove the requirement for components of tilting to account for the inclination data.

The declination anomalies are also potentially due to such tilting, but may also arise from rotations around vertical or inclined axes. The precise amount of structural tilting in these units is impossible to establish, due to the lack of palaeohorizontal markers. Standard palaeomagnetic tilt corrections involve restoring palaeohorizontal surfaces (e.g. bedding) back to horizontal by rotation around a line of strike, with remaining declination anomalies then being attributed to vertical axis rotations. Alternative approaches, such as the determination of net tectonic rotation axes [34] and inclination analyses [35], are preferred in complexly deformed regions where standard corrections can lead to potential declination errors. These methods still rely, however, on the identification of appropriate palaeohorizontal surfaces. It is possible to determine whether geologically realistic amounts of tilting can explain the observed declination anomalies, and hence whether components of vertical axis rotations are required, by modelling the effects of varying amounts of structural tilting on the revised Miocene reference direction. This approach has recently been used by Feinberg et al. [36] to interpret palaeomagnetic data from the Ronda peridotites of the Betic Cordillera, Spain. Fig. 6a shows the loci of remanence directions produced by tilting the reference direction in 10° increments up to 50°. Moderate amounts of tilting ($< 25°$) cannot completely explain the observed formation mean directions, regardless of the orientation of the axis of tilting. The minimum tilts which rotate the reference direction to the observed in situ directions are 51° about a strike of 182° for Mykonos and 36° about a strike of 356° for Naxos. These tilt axes are inconsistent with observed fault trends and recent structural models of Aegean core complexes. We

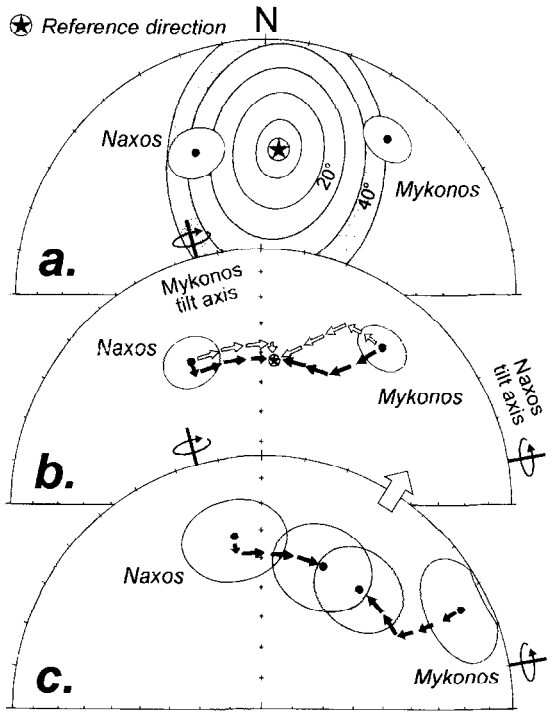


Fig. 6. (a) Stereoplot illustrating the effects of tilt about varying axes on the revised Miocene palaeomagnetic reference direction of $D = 005°$, $I = 42°$. Formation mean remanence directions and associated α_{95} cones of confidence are shown. These data can not be explained by moderate ($< 25°$) amounts of simple tilting, irrespective of the axis of tilting. (b) Rotation solutions (decompositions of net tectonic rotations) which are consistent with tilting about strikes of observed major block-bounding faults and which restore in situ magnetisations to the reference direction. Black arrows = tilt–rotation solution; white arrows = rotation–tilt solution. See text for explanation. (c) Mean AMS-derived extension directions and associated α_{95} cones of confidence (open ellipses) restored for the effects of tectonic rotations (filled ellipses) using the tilt–rotation solution. Large white arrow = suggested pre-rotation extension direction for the Cycladic Massif.

conclude that simple tilting alone can not account for the palaeomagnetic data.

In the absence of palaeohorizontal markers, an infinite number of net tectonic rotations can be found for each island which restore the observed in situ formation remanence direction back to the reference direction. The loci of possible poles of rotation are the great circle bisectrices between the formation mean directions and the reference direction. No unique pole of rotation can be found for either data set. Plausible sequential rotations can be proposed, however, by using additional data on the strike of the major normal faults which bound the sampled blocks. These faults control any tilting which may have occurred [37]. The procedure involves rotating the observed in situ remanence directions around horizontal axes parallel to the strikes of the bounding faults so that the inclinations are restored to that of the reference direction (42°). This defines the amounts of tilting about the fault strikes which are consistent with the palaeomagnetic data. Vertical axes rotations can then be applied to restore the tilt-corrected data to the reference declination (Fig. 6b). The resulting tilts and rotations which can account for the observed inclination flattenings and declination anomalies are:

Naxos:	tilt = 5° SSE, fault strike = 080°	rotation = 33° anticlockwise
Mykonos:	tilt = 27° SW, fault strike = 165°	rotation = 22° clockwise

It must be stressed that this solution is not unique. In particular, the order of tilting and vertical axis rotations is critical, with the opposite order giving:

Naxos:	rotation = 30° anticlockwise	tilt = 5° SSE
Mykonos:	rotation = 13° clockwise	tilt = 37° SW

It must be stressed again that these solutions represent only two possible decompositions of the net tectonic rotations required to restore remanences to the reference direction. Nevertheless, we consider these to be most consistent with observed regional structural trends. Irrespective of the precise method of decomposition, the observed palaeomagnetic data require the sampled intrusions to have undergone vertical axis rotations in opposite senses (Fig. 6b). Only the magnitudes of these rotations lack absolute constraints.

7. Restoration of extension directions

At face value, the AMS data for these intrusions (Table 1) imply widely divergent extension direc-

tions (Mykonos: 12/064; Naxos: 33/351). However, the remanence data suggest significant components of vertical axis rotations must be taken into account. Since intrusion was synchronous with extension, the extension directions derived from the AMS data must have been affected by these rotations. The corrections used to restore the remanence data to the Miocene reference direction are, therefore, also used to correct the magnetic lineation directions for each intrusion. Both the tilt–rotation and rotation–tilt solutions result in a marked convergence of these data after correction:

Tilt–rotation correction:	Mykonos: 38/040;	Naxos: 38/024 (Fig. 6c)
Rotation–tilt correction:	Mykonos: 45/041;	Naxos: 38/023

These restorations reconcile the apparently divergent Middle Miocene extension directions derived from both AMS and structural studies. Furthermore, the data imply that the true Middle Miocene extension direction in this region was nearly uniform along a NNE–SSW azimuth (Fig. 6c).

8. Wider implications and discussion

Our palaeomagnetic reconstruction of extension directions for exhumed lower–middle crustal rocks of the Cycladic Massif suggest little divergence of primary, Middle Miocene extension directions. We suggest that the observed divergence of extension directions in these rocks reflects subsequent differential block rotation rather than primary variations in the direction of ductile flow. Furthermore, the clockwise rotation observed on Mykonos is consistent with previously documented rotations in the internal Aegean zones further to the northwest. Kissel et al. [7] report a clockwise rotation of 45° from Middle Miocene volcanic rocks from near Kimi, Evvia (Fig. 1). Morris [9] documents a lower rotation angle of 15° from younger lacustrine sediments from the same area of Evvia, suggesting progressive post-Middle Miocene block rotation. This palaeomagnetic data set from upper crustal rocks at Kimi is also relevant to the problem of evaluating primary Miocene extension directions across the back-arc region. Using these data to correct observed ENE extension directions in the exhumed rocks of the metamorphic core complex on southern Evvia (Fig. 1) brings them into close alignment with the NNE–SSW directions determined for Mykonos and Naxos. We tentatively suggest that this agreement, obtained using palaeomagnetic data from upper crustal rocks to correct extension directions from lower–middle crustal rocks, is not merely coincidental but implies similar post-Middle Miocene rotation of both brittle and ductile portions of the crust. The rotation of ductile fabrics within rocks of the core complexes must have occurred after exhumation into the brittle field, since the magnetic remanence is partially carried by pyrrhotite, with a maximum unblocking temperature of 320°C.

Other Tertiary data from northern Greece show maximum clockwise rotations of a similar magnitude to those observed on Mykonos and Evvia ([10] and references therein). The available data therefore suggest comparable amounts of clockwise rotations within a domain extending from northern Greece as far south as Mykonos. The southern boundary of this domain must lie between Mykonos and Naxos, which is shown to have contrasting magnetic remanence, magnetic fabric and structural lineation directions, resulting from rotation in the opposite sense. This boundary corresponds to a major structural lineament identified in the straits between the islands by shallow seismic reflection experiments [38].

The model of Taymaz et al. [3] describes the active tectonics of the north and central Aegean by a system of broken, fault-bounded slats. These accommodate distributed WSW–ENE dextral shear between Turkey and Eurasia [1]. NW–SE trending slats along the eastern Aegean rotate in a clockwise sense, whereas ENE–WSW trending slats to the west rotate anticlockwise. This model, based primarily on seismological data, is believed to describe the active deformation of the region as far south as the volcanic arc in the Cyclades, and therefore encompasses the area of the present study. It is supported by recent geodetic observations which suggest that the crust in the Aegean is divided into a limited number of slowly deforming blocks separated by more rapidly deforming zones [39]. Our palaeomagnetic data and post-Middle Miocene structural reconstructions are

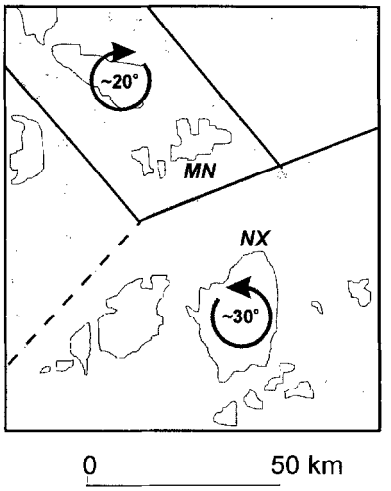


Fig. 7. Proposed location of Mykonos and Naxos within the broken slat model [3] for the Aegean. Mykonos and Naxos occupy positions on clockwise and anticlockwise rotating slats, respectively. The slat boundary corresponds to a major ENE–WSW trending fault between the islands, previously identified on shallow seismic sections. Major regional faults [4] are shown in half-tone (symbols as in Fig. 1); Black lines represent inferred slat boundaries; *MN* = Mykonos; *NX* = Naxos.

also compatible with this model, assuming that Mykonos occupies a position on one of the western slats and Naxos lies on an eastern slat (Fig. 7). The exact location within this model of the junction between opposing slats in the central Aegean is uncertain, but our analyses suggest that one such boundary corresponds with the ENE–WSW trending structure separating Mykonos and Naxos, identified as a major crustal shear zone from shallow seismic surveys [38]. The potential relationship between this structure and the broken slat model [3] was noted by Gautier and Brun [4]. However, these authors preferred to interpret the divergence of regional extension directions across this fault as a consequence of primary radial spreading rather than as a result of fault block rotation. In support of this conclusion, they noted an apparent consistency between *uncorrected* Miocene extension directions observed onshore in the Cyclades and Pliocene extension directions inferred from the geometry of offshore faults. In the light of the new palaeomagnetic data reported in the present study, several aspects of these interpretations may require re-evaluation. In particular, after correcting for the effects of block rotation, we find little evidence that the direction of post-Middle Miocene ductile extension in the Cycladic Massif was originally divergent. It is worth noting, however, that our *corrected* Miocene extension directions correspond more readily with the inferred Pliocene–Quaternary extension direction for this region [11]. Further palaeomagnetic data from the Cyclades are now needed to determine whether or not the primary, Miocene direction of extension was uniform across the entire back-arc region.

Palaeomagnetic data from both islands suggest finite opposing rotations of the order of 20–30°. A maximum age for the initiation of rotation is provided by radiometric cooling ages, which suggest the intrusions entered the brittle regime at ca. 10 Ma. [40]. Instantaneous rotation rates for the Cycladic Massif, inferred from the broken slat model [3], vary from 5.6 to 7.2°/Ma for the western slats and from 2.8 to 3.6°/Ma for the eastern slats. Assuming that these rotation rates can be extrapolated through time, a minimum period of 3–4 Ma and 9–12 Ma is required to achieve the rotations observed on Mykonos and Naxos, respectively.

Finally, this study highlights the usefulness of the palaeomagnetic technique in tectonic studies, even in settings where structural tilt corrections are difficult to apply. Tectonic models based on observed structural data from complexly deformed regions are potentially in error unless tectonic rotations documented through palaeomagnetic studies are taken into account.

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